

MATHEMATICS

MATHEMATICS GRADE 12 TERM 2 CONTROLLED ASSESSMENT PACK

SUBJECT	MATHEMATICS
GRADE	12
ASSESSMENT TASK	MATHEMATICS GRADE 12 TERM 2 CONTROLLED ASSESSMENT PACK
MARKS	50
TIME	1 hour

Educator approval required before classroom use.

INSTRUCTIONS AND INFORMATION

- 1. Read all instructions carefully.
- 2. Answer all questions or complete all activities as instructed.
- 3. Show all working where calculations, planning, diagrams or explanations are required.
- 4. Use the space provided by the educator, answer booklet or learner task book.
- 5. Write neatly and clearly.
- 6. Do not use a calculator unless the educator allows it.

VISUAL MATERIAL

FIGURE 1: Working Grid And Method Marks

Mathematics		CALCULATION WORKSPACE
Mathematics Grade 12 Term 2 Controlled Assessment Pack		
Working grid and method marks		
1.1 (6)	Solve a routine problem linked to the stated term focus and show all algebraic working.	
1.2 (6)	Interpret the supplied graph, table or diagram and state the meaning of one key feature.	
1.3 (8)	Complete a multi-step calculation using the supplied values and units.	
2.1 (10)	Set up and solve a non-routine problem from the term context.	
Award method marks from visible working before final-answer marks.		

QUESTIONS

QUESTION 1: SECTION A: CONCEPTS AND PROCEDURES

Instructions: Show all working.

Stimulus: Topic focus: calculus, rates of change, analytical geometry and trigonometry.

1.1. Solve a routine problem linked to the stated term focus and show all algebraic working.
(6)

1.2. Interpret the supplied graph, table or diagram and state the meaning of one key feature. **(6)**

1.3. Complete a multi-step calculation using the supplied values and units. **(8)**

QUESTION 2: SECTION B: PROBLEM SOLVING

Instructions: Use the workspace and justify each step.

Stimulus: Problem context: derivatives, tangents, optimisation and coordinate reasoning.

2.1. Set up and solve a non-routine problem from the term context. **(10)**

2.2. Justify one method step or proof statement. **(5)**

QUESTION 3: SECTION C: INTEGRATED APPLICATION

Instructions: Combine concepts from the term pack.

Stimulus: Integrated focus: integrated calculation problem with graph or diagram interpretation.

3.1. Answer an integrated application question and comment on the reasonableness of the result. **(15)**

MARKING GUIDELINE

TASK 1: SECTION A: CONCEPTS AND PROCEDURES

1.1. Solve a routine problem linked to the stated term focus and show all algebraic working. **(6)**

- Correct method.
- Accurate simplification.
- Final answer stated.

1.2. Interpret the supplied graph, table or diagram and state the meaning of one key feature. **(6)**

- Feature identified.
- Mathematical meaning explained.

1.3. Complete a multi-step calculation using the supplied values and units. **(8)**

- Values substituted.
- Steps logical.
- Answer with unit.

TASK 2: SECTION B: PROBLEM SOLVING

2.1. Set up and solve a non-routine problem from the term context. **(10)**

- Correct representation.
- Valid strategy.
- Accurate conclusion.

2.2. Justify one method step or proof statement. **(5)**

- Relevant theorem/rule.
- Reasoning clear.

TASK 3: SECTION C: INTEGRATED APPLICATION

3.1. Answer an integrated application question and comment on the reasonableness of the result. **(15)**

- Correct plan.
- Accurate working.
- Reasonableness comment.

RUBRIC

Criterion	Marks	Descriptor
Subject accuracy and terminology	12	Uses accurate concepts and terminology.
Evidence, data or source	12	Uses supplied evidence

use		appropriately.
Reasoning and explanation	12	Shows logical, grade-appropriate reasoning.
Communication and completion	14	Answers are clear, complete and structured.

EDUCATOR REVIEW CHECKLIST

- Confirm CAPS term and topic fit before classroom use.
- Insert or approve final source material where the pack requests educator-supplied sources.
- Check mark allocation, memo wording and learner level.
- Confirm visuals are suitable and not treated as official source material unless sourced separately.